



PERGAMON

Atmospheric Environment 35 (2001) 579–587

**ATMOSPHERIC
ENVIRONMENT**

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Minimizing the effect of automotive pollution in urban geometry using mathematical optimization

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Received 15 November 1999; accepted 30 May 2000

Abstract

One of the factors that needs to be considered during the layout of new urban geometry (e.g. street direction, spacing and width, building height restrictions) is the effect of the air pollution associated with the automotive transport that would use routes in this urban area. Although the pollution is generated at street level, its effect can be widespread due to interaction of the pollutant dispersion and diffusion with the wind speed and direction. In order to study the effect of a new urban geometry on the pollutant levels and dispersion, a very time-consuming experimental or parametric numerical study would have to be performed. This paper proposes an alternative approach, that of combining mathematical optimization with the techniques of computational fluid dynamics (CFD). In essence, the meteorological information as represented by a wind rose (wind speed and direction), is used to calculate pollutant levels as a function of urban geometry variables: street canyon depth and street canyon width. The pollutant source specified in conjunction with a traffic scenario with CO is used as pollutant. The main aim of the study is to be able to suggest the most beneficial configuration of an idealized urban geometry that minimizes the peak pollutant levels due to assumed traffic distributions. This study uses two mathematical optimization methods. The first method is implemented through a successive maximization–minimization approach, while the second method determines the location of saddle points of the pollutant level, considered as a function of urban geometry and wind rose. Locally, a saddle point gives the best urban geometry for the worst meteorological scenario. The commercial CFD code, STAR-CD, is coupled with a version of the DYNAMIC-Q optimization algorithm of Snyman, first to successively locate maxima and minima in a min–max approach; and then to locate saddle points. It is shown that the saddle-point method is more cost-effective. The methodology presented in this paper can readily be extended to optimize traffic patterns for existing geometry or in the development of geometry modification for pollution control or toxic releases. © 2000 Elsevier Science Ltd. All rights reserved.

Keywords: Automotive pollution; Computational fluid dynamics; Mathematical optimization; Saddle point

1. Introduction

There is an increasing interest in the contribution of automotive pollution sources to urban air pollution. This is mainly due to the large proportion of urban pollution

due to transport-related sources, e.g., in Greater London and Berlin in 1993, the carbon monoxide (CO) from road transport sources contributed to approximately 57% of the total CO budget for these cities (Seika et al., 1998). Extensive emission inventory studies are being performed worldwide to quantify the sources due to road transport. The modeling of the impact of automotive pollutants has also been receiving increased attention (e.g. Albergel and Jasmin, 1998; Sini et al., 1996). Included in this modeling, is the determination of the

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flow patterns in urban geometry due to local effects as well as meso-scale meteorological conditions. With ever-increasing computer power, computational fluid dynamics (CFD) techniques have been increasingly applied to the simulation of atmospheric problems. For the design of regulations to govern new developments and changes to existing urban environments, advance simulation of conditions is required. This typically leads to a design problem with many design variables. Amongst these variables are traffic flow parameters (speed, direction, type of vehicles, age of fleet, etc.), street canyon depth and width (linked to building height restrictions), and street spacing and orientation, to name a few. CFD can play an important role in the analysis of the impact of automotive pollution on urban pollution levels and distribution. However, one can imagine that an extensive parametric variation of the design variables mentioned would result in an impractical situation as far as the required computing overheads are concerned. This is exacerbated by the fact that the grid resolution and level of modeling accuracy required for atmospheric flows lead to very computing-intensive CFD models.

A more efficient approach is to combine CFD with mathematical optimization, in order to provide an optimum solution after a number of design iterations that are significantly less than would be required during a parametric study of the design variables. Moreover, mathematical optimization automatically takes the cross-influence of the variables into account. Previous work by the authors (Craig et al., 1999a) illustrated this approach for pollutants due to stacks. That particular study considered a two-dimensional urban geometry in the form of a street canyon and determined the optimal placement and height of a stack (and implied pollutant-emitting factory) relative to a street canyon with a constraint being placed on the pollutant level in the canyon. Additional design variables considered dealt with the street canyon width and depth, as well as with the specification of a barrier that attempted to deflect the stack plume as it approached the street canyon.

The current study concentrates on the dispersion of automotive pollution as emitted at street level in an idealized geometry due to meteorological conditions. In essence, meteorological information as represented by a wind rose (wind speed and direction), is used to calculate pollutant levels as a function of the urban geometry variables, street canyon depth and street canyon width. The pollutant source is specified for a typical rush-hour traffic scenario with CO used as pollutant. The choice of idealized geometry was governed by three items. First, it had to allow for 3D effects, to progress from the previous work by the authors (Craig et al., 1999a). Second, the geometry had to be symmetric in order to reduce the range of wind directions considered. Third, the number

of design variables (parameters defining the geometry and meteorological conditions) were kept to a minimum to reduce the computational time.

The study has two aims. The first aim is to be able to suggest the most beneficial configuration of urban geometry that minimizes the integrated pollutant levels due to the assumed traffic distributions. Of course, the optimum found is only as accurate as the accuracy of the numerical simulation model used, and the applicability of the optimum result is limited by the idealized nature of the test geometry. The second aim of the study is the utilization of an illustrative problem to introduce a special and new optimization methodology to handle min-max problems economically. A recent study (Craig and De Kock, 1999) separated the problem into a maximization problem (finding the wind direction and speed that produces the highest peak pollutant level at pedestrian height for a given urban geometry), and a minimization problem (modifying the geometry in order to reduce the peak pollutant level), and successively repeating these two problems to convergence. This first method will be labeled *Method 1* in the paper. A continuation of this study, presented in part in Craig et al. (2000a), varies this approach by combining the maximization and minimization problems into a single search for a stationary point, which corresponds to a saddle point, of the pollutant level considered simultaneously as a separable function of urban geometry and wind direction and speed. Essentially, the proposed search procedure seeks the saddle point that corresponds to the minimum, with respect to the urban geometry, of the maximum with respect to the wind rose of the pollutant level (i.e., a min-max problem is solved). This second method will be labeled *Method 2* in the paper. The paper outlines both these two methods and discusses their relative merits.

The pollutant level for a given geometry and wind rose is computed by means of the commercial CFD code, STAR-CD (1998). This code is coupled with a version of the DYNAMIC-Q optimization algorithm of Snyman (Snyman, 1982, 1983; Snyman et al., 1994), used first to successively locate a maximum with respect to wind-rose conditions, followed by the determination of a minimum with respect to geometry (Method 1); and then adapted to find saddle points instead of strict minima or maxima (Method 2). The DYNAMIC-Q methodology has been proven to be robust in a variety of thermoflow applications (Craig et al., 1999a,b,c; Craig and Venter, 1999; De Kock et al., 2000).

The next section will define the optimization problem, where after the theoretical modeling employed will be described. This includes CFD and mathematical optimization modeling of the air pollution problem. This section is followed by a presentation of the results for the two optimization methods, with conclusions made from the results concluding the paper.

2. Problem definition and formulation

The idealized urban automotive air pollution optimization problem considered for illustrative purposes in this study is depicted graphically in Fig. 1. Shown is a 3×3 array of buildings with two streets in each of the two main directions. The buildings are square-shaped when viewed from above, and the streets are assumed to have a constant width with uniform spacing. The design variables considered as a first step are street width and building height. The size of the urban section considered ($500 \text{ m} \times 500 \text{ m}$), as well as the up- and downwind fetch of 5 km, remain constant for this initial study. The height of the solution domain is 300 m, with an inviscid (zero velocity gradient) boundary condition being applied on the upper boundary. This is obviously an idealized urban geometry, but at least allows for an extension of two-dimensional street canyon studies as realistic three-dimensional flow phenomena are possible due to the wind direction variable.

CO is used as a representative automotive emission since it is relatively stable, easily measured and comes mainly from vehicular emissions. CO has also been used as proxy for other pollutants, including NO_x , hydrocarbons and PM10 (Penn and Croxford, 1998). The CO values due to vehicles typically range from 1 g/km (Samaras et al., 1998) for cars running hot with catalytic converters, to 80 g/km (Spadaro et al., 1998) for conventional cars without converters in cold-start short distance cycles. In typical rush-hour conditions, there is a mix in the type of vehicles present in the fleet. Albergel and Jasmin (1998) use emissions based on INRETS and CORINAIR reports for passenger cars and light duty vehicles in evening rush-hour traffic at 7.3 km/h. The CO value they derive from these reports is 30.57 mg/s, resulting in about 60 g/km for a 4-lane avenue. A value of 15 g/km CO emission per vehicle lane is used in this study as representative, but could easily be varied due to the

method of implementation. The number of lanes is calculated from the street width suggested by the optimizer, and is a real value. The actual CO emission source is then calculated from this real value.

The two urban geometry design variables are shown in Fig. 1. $x_1 = w$ is the street width and $x_2 = h$ is the building height, i.e. $\mathbf{x} = (x_1, x_2)$. Due to the symmetric lay-out of the geometry, the range of wind directions that needs to be considered is only one quarter of all wind directions, i.e. from 0 to 45° in Fig. 1. The atmosphere is considered as neutrally stable in this study. The wind profile is assumed to be steady and to follow a power law distribution. In general, pollutant episodes with transient wind conditions can be considered if instantaneous pollutant levels are required. The current type of analysis is more suited for averaged or accumulative pollutant scenarios. The wind direction, α and wind speed V are variables that affect the pollutant level and are allowed to vary continuously between specified limits. The aim is to find values of the variables $y_1 = \alpha$ and $y_2 = V$, i.e., $\mathbf{y} = (y_1, y_2)$, that would produce the worst-case scenario as far as CO pollutant concentration level is considered, and then minimizing this level by varying the geometrical variables, $\mathbf{x}$.

The resulting optimization problem can be seen as a nested optimization problem (i.e., minimize $\left\{ \begin{matrix} \text{maximum } f(\mathbf{x}, \mathbf{y}) \\ \text{w.r.t. } \mathbf{x} \end{matrix} \right\}$ with $f(\mathbf{x}, \mathbf{y}) = \text{Con}(\mathbf{x}, \mathbf{y})$ the peak pollutant level at pedestrian height. This problem can be treated in two ways. The first is a sequential maximization and minimization approach that can be formulated as follows:

Method 1: Successive maximization and minimization:

$$\text{minimize}_{\mathbf{x}=[h,w]} \left\{ \begin{matrix} \text{maximum } \text{Con}(\mathbf{x}, \mathbf{y}) \\ \mathbf{y}=(\alpha, V) \end{matrix} \right\}, \quad (1)$$

where the upper and lower limits on the variation of the variables $\mathbf{x}$ and $\mathbf{y}$ are given by $x_j^{\min}$ and $x_j^{\max}$ and $y_j^{\min}$ and $y_j^{\max}$, respectively.

In the second method, the fact is recognized that the saddle point $(\mathbf{x}^*, \mathbf{y}^*)$ of the pollutant level $f(\mathbf{x}, \mathbf{y})$, considered as a simultaneous and separable function of $\mathbf{x}$ and $\mathbf{y}$, corresponds to the solution of the above min-max problem (1). Here the *saddle-point problem* may be mathematically formulated as follows.

Method 2: Saddle-point optimization: Determine $(\mathbf{x}^*, \mathbf{y}^*)$ such that

$$f(\mathbf{x}, \mathbf{y}^*) \geq f(\mathbf{x}^*, \mathbf{y}^*) \geq f(\mathbf{x}^*, \mathbf{y}) \quad (2)$$

subject to

$$x_j^{\min} \leq x_j \leq x_j^{\max}, \quad j = 1, 2,$$

$$y_j^{\min} \leq y_j \leq y_j^{\max}, \quad j = 1, 2.$$

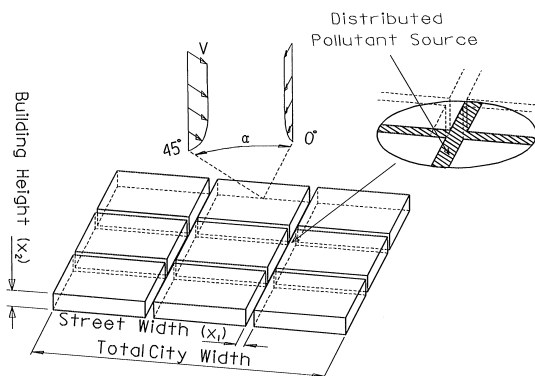


Fig. 1. Graphical representation of automotive urban air pollution problem.

In both methods, the objective function, $f(\mathbf{x}, \mathbf{y}) = \text{Con}(\mathbf{x}, \mathbf{y})$ is the CO level obtained from a CFD simulation given a specific geometry (defined by $\mathbf{x}$) at a 2 m level in the streets for the specified wind speed and direction (defined by $\mathbf{y}$).

3. Theoretical modeling

3.1. CFD modeling

3.1.1. Grid generation

The grid is generated using algebraic grid generation techniques (transfinite interpolation and Vinokur stretching (Vinokur, 1983) exterior to the CFD code. A new grid is generated for each perturbation of the design variables. The total flow domain is $5 \text{ km} \times 5 \text{ km}$ square and 300 m high. Grid sizes used varied from 200,000 to 300,000 cells.

3.1.2. Flow solver

The commercial CFD code, STAR-CD is used, as mentioned above. The Reynolds-averaged Navier–Stokes equations are solved with turbulence closure provided by the standard $k-\epsilon$ turbulence model. The SIMPLE algorithm that uses a pressure–velocity correction approach to iteratively solve for the Navier–Stokes equations, was employed. The simulations were performed on an Aspen Durango with DEC Alpha 600 MHz processor. When being the sole user on this machine, a run-time of approximately 0.15 ms/grid point/iteration was achieved. This translates to each iteration in Method 1 taking approximately 13 h. This extreme computational demand underlines the need for an economic optimization procedure.

3.1.3. Flow solutions

To illustrate the type of solutions that are used in the optimization process, a representative case is given in Fig. 2. Shown are the steady-state velocity vectors for an oblique wind condition of $\alpha = 20^\circ$. The vectors displayed are at the 2 m elevation. Note that the wind speed and direction in the street canyons differ from the prevailing wind. Of interest is the level of ‘washing-out’ that occurs in the streets due to the wind, as it impacts the dispersion of automotive emissions.

3.1.4. CO concentrations

To illustrate the types of CO concentration distributions that are computed in the street canyons due to the types of wind distribution shown in Fig. 2 above, CO concentration levels are shown at the 2 m elevation in Fig. 3. In Fig. 3(a), the wind direction is head on ($\alpha = 0^\circ$) while in Fig. 3(b), ($\alpha = 20^\circ$). Note the complex distribution of CO as exhibited by the close-ups of the street

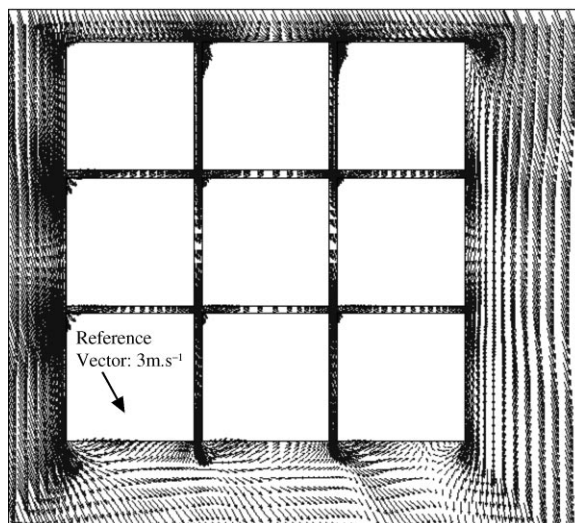


Fig. 2. Velocity vectors at 2 m elevation. Wind direction, $y_1 = \alpha = 20^\circ$; wind speed, $y_2 = V = 3 \text{ m s}^{-1}$; CO source: 15 g/km/lane ; design variables: $x_1 = w = 10 \text{ m}$, $x_2 = h = 15 \text{ m}$.

intersections. Note that the highest CO level of the two wind conditions shown occurs in the ($\alpha = 0^\circ$) case.

Fig. 4 shows the streamlines and CO contours at a street intersection for the ($\alpha = 0^\circ$) case. The image was generated in VR-FloViz (Craig et al., 1999d), a post-processor utilizing virtual reality. Note how the CO accumulates in pockets at the elevation shown. The wind (blowing from right to left in the figure) washes out the street canyons aligned with the wind, but complex vortical patterns can be seen to develop in the cross-direction street canyons, resulting in the accumulation of CO mentioned.

3.2. Mathematical optimization

3.2.1. Optimization algorithm

The reader is referred to Craig et al. (1999a, b, c, d) for the detail implementation of the optimization method used in this study. In essence, the DYNAMIC-Q method of Snyman et al. (1994) involves the application of a dynamic trajectory method for unconstrained optimization (Snyman, 1982, 1983), adapted to handle constrained problems through appropriate penalty function formulations. This DYNAMIC method is applied to successive approximate quadratic subproblems that are constructed from sampling, at relatively high computational expense, the behavior of the objective function (described in (1,2)) at successive approximate solution points in the design space. The subproblems, which are analytically simple, are solved quickly and economically using the adapted dynamic trajectory method. In Method 2, the method is modified to find saddle points by switching the sign of the

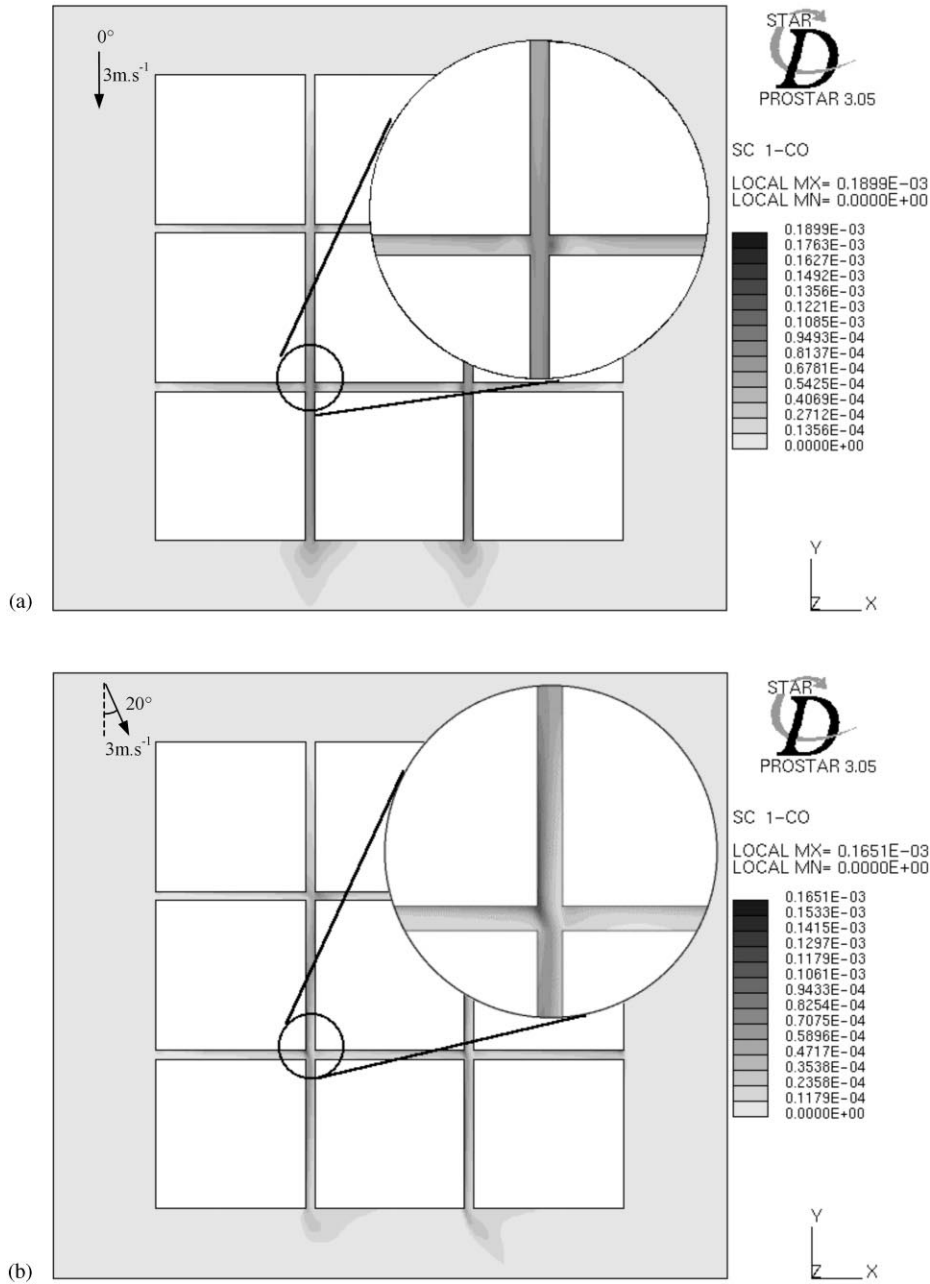


Fig. 3. (a). CO concentration contours at 2m elevation. Wind direction, $y_1 = \alpha = 0^\circ$; wind speed, $y_2 = V = 3 \text{ m s}^{-1}$; CO source: 15 g/km/lane ; design variables: $x_1 = w = 10 \text{ m}$, $x_2 = h = 15 \text{ m}$ (b). CO concentration contours at 2m elevation. Wind direction, $y_1 = \alpha = 20^\circ$; wind speed, $y_2 = V = 3 \text{ m s}^{-1}$; CO source: 15 g/km/lane ; design variables: $x_1 = w = 10 \text{ m}$, $x_2 = h = 15 \text{ m}$.

relevant components of the objective function gradient vector. This modification is outlined in the next section.

3.2.2. Saddle-point optimization

To illustrate the implementation of DYNAMIC-Q in the determination of saddle points, consider the uncon-

strained problem

$$\underset{\text{w.r.t. } \mathbf{x}}{\text{minimize}} \left\{ \underset{\text{w.r.t. } \mathbf{y}}{\text{maximum}} f(\mathbf{x}, \mathbf{y}) \right\}, \mathbf{x} \in R^p, \mathbf{y} \in R^q. \quad (3)$$

Assume that the solution lies at the saddle point $(\mathbf{x}^*, \mathbf{y}^*)$, and that f is convex with respect to $\mathbf{x}$, and concave with

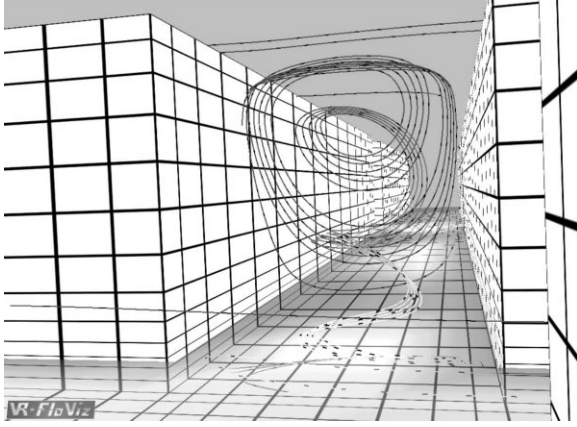


Fig. 4. Streamlines and CO concentration contours. Wind direction, $y_1 = \alpha = 0^\circ$; wind speed, $y_2 = V = 3 \text{ m s}^{-1}$; CO source: 15 g/km/lane ; design variables: $x_1 = w = 10 \text{ m}$, $x_2 = h = 15 \text{ m}$.

respect to y . If these two sets of variables are separable (i.e., $f(\mathbf{z}) = f(\mathbf{x}, \mathbf{y}) = f_1(\mathbf{x}) + f_2(\mathbf{y})$, so that they do not have a significant cross-influence on each other), then this problem becomes tractable as outlined below.

If the values of $f(\mathbf{x}, \mathbf{y})$ and the associated gradient vectors are known at two successive design points $(\mathbf{x}^{(k-1)}, \mathbf{y}^{(k-1)})$ and $(\mathbf{x}^{(k)}, \mathbf{y}^{(k)})$, then the spherical quadratic approximation to the objective function is taken as

$$\tilde{f}(\mathbf{x}, \mathbf{y}) = f(\mathbf{x}^{(k)}, \mathbf{y}^{(k)}) + \nabla_{\mathbf{x}}^T f^k(\mathbf{x} - \mathbf{x}^{(k)}) + \nabla_{\mathbf{y}}^T f^k(\mathbf{y} - \mathbf{y}^{(k)}) + \frac{1}{2}a^{(k)}\|\mathbf{x} - \mathbf{x}^{(k)}\|^2 - \frac{1}{2}a^{(k)}\|\mathbf{y} - \mathbf{y}^{(k)}\|^2 \quad (4)$$

with $a^{(k)}$ given by

$$a^{(k)} = \frac{2\{f^{k-1} - f^k - \nabla_{\mathbf{x}}^T f^k(\mathbf{x}^{(k-1)} - \mathbf{x}^{(k)}) - \nabla_{\mathbf{y}}^T f^k(\mathbf{y}^{(k-1)} - \mathbf{y}^{(k)})\}}{\{\|\mathbf{x}^{(k-1)} - \mathbf{x}^{(k)}\|^2 - \|\mathbf{y}^{(k-1)} - \mathbf{y}^{(k)}\|^2\}},$$

where $f^k = f(\mathbf{x}^{(k)}, \mathbf{y}^{(k)})$.

For application in DYNAMIC-Q, the saddle point of a fictitious function $\text{fic}(\mathbf{x}, \mathbf{y})$ is found if the gradients in the dynamic trajectory algorithm are chosen as follows:

$$\nabla_{\mathbf{x}} \text{fic} = \nabla_{\mathbf{x}} f^k + a^{(k)}(\mathbf{x} - \mathbf{x}^{(k)}) \quad (5)$$

and

$$\nabla_{\mathbf{y}} \text{fic} = -\nabla_{\mathbf{y}} f^k + a^{(k)}(\mathbf{y} - \mathbf{y}^{(k)}). \quad (6)$$

By switching the sign of the actual gradient with respect to $\mathbf{y}$ of $\tilde{f}$ in (4), as shown in the latter expression (6), the algorithm, since it uses only gradient information, effectively minimizes a 'fictitious' convex function $\text{fic}(\mathbf{x}, \mathbf{y})$, the minimum of which $(\mathbf{x}^{(k+1)}, \mathbf{y}^{(k+1)})$ corresponds to the saddle point of $\tilde{f}(\mathbf{x}, \mathbf{y})$.

The computed function and gradient values at the point $\mathbf{z}^{(k+1)} = (\mathbf{x}^{(k+1)}, \mathbf{y}^{(k+1)})$ may now be used to construct the next approximation $\tilde{f}(\mathbf{x}, \mathbf{y})$ to $f(\mathbf{x}, \mathbf{y})$, as given by (4) but for $(\mathbf{x}^{(k+1)}, \mathbf{y}^{(k+1)})$, and the approximation procedure is continued until convergence is obtained to $(\mathbf{x}^*, \mathbf{y}^*)$. It is further assumed that this procedure, as applied to the objective function above, is also valid in the presence of constraints with, of course, no such adjustment ((5) and (6)) applied to the gradients of the constraint functions.

3.2.3. Selection of objective function

The selection of the objective function used in this study deserves some mention. The allowable levels due to pollutant sources are determined by legislation and depend on the sources. For the current implementation, the mathematical nature of the pollutant distribution must also be taken into account. Possible alternatives are e.g. the maximum CO concentration level in the street canyon volume, the maximum level at a specified height (e.g. 2 m), the integrated CO concentration level in the street canyon volume, or the integrated CO concentration level at the 2 m height. Alternatively, the total CO emitted (kg) that remains in the street canyon volume during steady-state conditions can be used, preferably normalized by the street canyon volume for valid comparison for different street canyon parameters. The $\text{Con}(\mathbf{x}, \mathbf{y})$ used in the objective function (1, 2) in the current study is that of the maximum CO concentration level at the 2 m elevation.

For more realistic scenarios, the choice of objective function would typically be driven by the objective of the study (instantaneous release, steady release, point sources, surface/volume sources, different episodes, etc.). As the choice of objective function is such an integral part of

the formulation of the optimization problem, considerable experience is required for an objective function that would allow convergence to an optimum. The results section below provides some more information on the behaviour of possible objective functions for the current problem.

3.2.4. Combination of CFD and mathematical optimization

The combination is performed in the Unix environment using FORTRAN 90, which allows dynamic memory allocation and system calls. This integration of the CFD and mathematical optimization algorithm into one program allows for an automated optimization process and a quick turnaround when formulating and implementing new optimization problems. Recently, this design optimization process has been implemented into

a graphical user interface environment for personal computers (Craig et al., 2000b).

4. Results and discussion

Table 1 shows the limits on the design variables used as well as the model parameters used for both methods. z_0 is the reference height of the power law wind profile, with n the power law coefficient. The Schmidt number, which relates the mass transfer due to diffusion to the mass transfer due to momentum of the CO, together with the molecular diffusivity, specify the convection and diffusion of the CO transport in the CFD model. The initial design used in both cases is that shown in Figs. 2 and 3(b), i.e. wind direction, $y_1 = \alpha = 20^\circ$; wind speed, $y_2 = V = 3 \text{ m s}^{-1}$; $x_1 = w = 10 \text{ m}$ and $x_2 = h = 15 \text{ m}$.

Method 1: Successive maximization and minimization: The histories of the objective function as well as that of the design variables for Method 1 are given in Fig. 5. The optimization was not allowed to go to full convergence in the intermediate (nested) loops to speed up the overall convergence rate. Note that the wind speed is reduced to the minimum allowable for the worst-case meteorological condition (iterations 1–4), as expected, but, that the worst wind angles of 17.319° and 5.569° for the two building height and street width configurations shown, could not have been predicted by any other method. Given the first worst-case wind condition (iteration 4), the optimizer then proceeds to reduce the building height and increase the street width (iterations 5–9) in order to reduce the maximum CO level at the 2 m elevation. Surprisingly, the optimum obtained does not represent the widest street width allowed, but the value shown in the figure caption for iteration 9. In the next section, shown in Fig. 5, the optimizer again performs a maximization problem (iterations 10–13) on the wind conditions with the building height and street width values obtained in iteration 9. With the lower buildings and wider streets of iteration 9, the maximum concentration at iteration 13 is lower than the first worst case (iteration 4), indicating that the optimizer is moving in the right direction. In the final minimization problem shown (iterations 14–17), the CO level is reduced slightly while the street width is reduced to 7.8 m and the building height increased to 8 m.

Method 2: Saddle-point optimization: The histories of the objective function as well as that of the design

variables for Method 2 are given in Fig. 6. It can be seen that the optimization process converges in only six steps as opposed to the 17 iterations that were required in Method 1. In addition, the final CO value is lower than that obtained in Method 1. The location of the CO concentration peak at the 2 m elevation for each optimization iteration, is indicated in Fig. 7. Note the final location (iteration 6) is located in a street 'aligned' with the wind direction (10.9°). This fact stresses the requirement for a detailed wind field solution when studying pollution transport, as separation regions caused by sharp corners of buildings can lead to non-intuitive locations of high local concentration. Note that the solution obtained in iteration 6 may only be a local minimum, but that a multiple starting strategy or some other global optimization approach would be required to determine whether this is indeed the case. If the variable sets

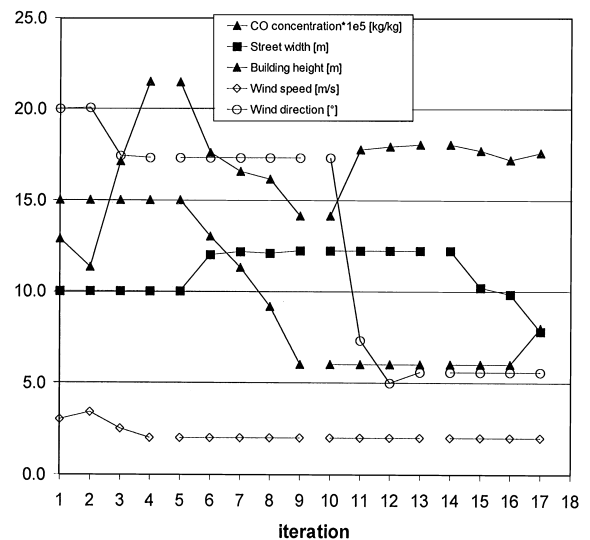


Fig. 5. History of CO concentration, wind speed and direction, and street width and building height. CO source: 15 g/km/lane (Method 1). Worst-case(1): Iteration 4: wind direction, $y_1 = \alpha = 17.319^\circ$; wind speed, $y_2 = V = 2.000 \text{ m s}^{-1}$, CO = 21.50e – 5 kg/kg Optimized geometry: Iteration 9: $x_1 = w = 12.213 \text{ m}$, $x_2 = h = 6.000 \text{ m}$, CO = 14.12e – 5 kg/kg. Worst-case (2): Iteration 13: wind direction, $y_1 = \alpha = 5.569^\circ$; wind speed, $y_2 = V = 2.000 \text{ m s}^{-1}$, CO = 18.06 e – 5 kg/kg Optimized geometry: Iteration 17: $x_1 = w = 7.8327 \text{ m}$, $x_2 = h = 8.000 \text{ m}$, CO = 17.60e – 5 kg/kg.

Table 1
Parameters used for optimization and limits on design variables

Street width $x_1 = w$ (m)	Building height $x_2 = h$ (m)	Wind direction $y_1 = \alpha$ (°)	Wind speed $y_2 = V$ (m s ⁻¹)	Power law z_0, n (m, –)	Schmidt number	Diffusivity (m ² s ⁻¹)
5–14	6–30	0–45	2–5	10, 0.18	0.77	2.02×10^{-5}

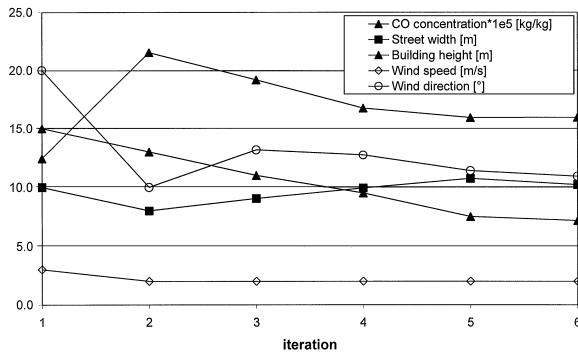


Fig. 6. History of CO concentration, wind speed and direction, and street width and building height. CO source: 15 g/km/lane (Method 2). Iteration 6: wind direction, $y_1 = \alpha = 10.916^\circ$; wind speed, $y_2 = V = 2.000 \text{ m s}^{-1}$, CO = $15.94 \times 10^{-5} \text{ kg/kg}$. Optimized geometry: $x_1 = w = 10.2 \text{ m}$, $x_2 = h = 7.156 \text{ m}$.

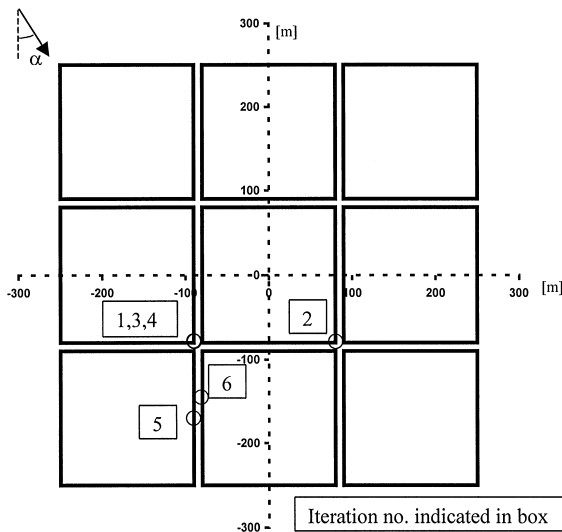


Fig. 7. Location of maximum CO concentration at 2m elevation for each optimization iteration. (relative wind direction indicated on figure). CO source: 15 g/km/lane (Method 2).

(x and y) had not been separable, this method would not have converged, unless a transformation had been applied that will require a knowledge of the eigenvalues of the Hessian of the objective function, f , which are not readily available.

Two of the other objective function candidates mentioned above (maximum CO concentration level in the street canyon volume and integrated CO concentration level in the street canyon volume) were also monitored during the saddle-point optimization process. Their history is depicted in Fig. 8 together with the chosen objective function of maximum CO concentration level at the 2m elevation (previously shown in Fig. 6). Interestingly,

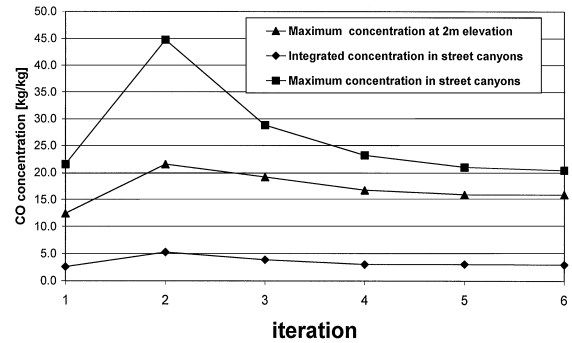


Fig. 8. History of CO concentration: peak value at 2m elevation (objective function); average value in street canyons; peak value in street canyons. CO source: 15 g/km/lane (Method 2).

they both exhibit the same trend as the chosen objective function, with the overall maximum CO level being higher as expected (located closer to the CO source – the traffic), and the averaged or integrated CO level being lower, as expected.

In summary, note that the solution for both methods in essence represents four solutions of wind direction due to the symmetry in the problem.

5. Conclusion

This study illustrates the combination of CFD and mathematical optimization to the minimization of the effect of automotive pollutants in urban environments. Although an idealized geometry is investigated, the method is readily extendable to more complex cases, including toxic releases of pollutants. In a realistic (real-world) application of this method, one would e.g. apply the method to a real urban scenario and optimize the modification of the geometry (through the specification of appropriate parameters) based on pollution sources and randomly varying meteorological conditions (again parameterized). In such an application, the variation of atmospheric stability conditions, by including e.g. a very stable lower atmosphere that inhibits the dilution of microscale pollutants such as CO, can also be parameterized and included in the optimization problem. In the CFD model, the variation of atmospheric stability would be affected through the specification of the gravitational field properties and temperature boundary conditions.

The optimization algorithm proved to be robust in the cases presented. It was shown that the saddle-point optimization method required a significantly reduced number of iterations to converge to an optimum when compared to the successive maximization–minimization method that might intuitively be used for problems of this type. The saddle-point method has great potential in

optimization problems where input or environment variables are stochastic by nature. With this is meant problems where a feasible design is required for a variety of (a priori) unknown conditions. This should not be confused with optimal control systems where the design 'follows' the input variables. The field of application of the current methodology would be where the design or 'plant' is finalized before operation and remains fixed. Examples of applications include the design of escape routes during toxic release episodes through the implementation of geometric barriers, and the optimization of traffic flow patterns for all possible or measured meteorological conditions.

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